WORKED SOLUTION

STEP II 1994, Question 11

Mechanics • Projectiles and Explosions

STEP II 1994, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/94-s2-q11

The key idea

The explosion happens at a known height, the top of the vertical flight, and at that moment the shell is at rest. So the two fragments start from the same point with equal and opposite velocities. If we describe the direction by one angle θ, each landing time comes from a quadratic, and the gap between the landing points becomes a single function of θ. The two cases arise because the maximum of that function lies either at an end of the allowed range or at an interior turning point.

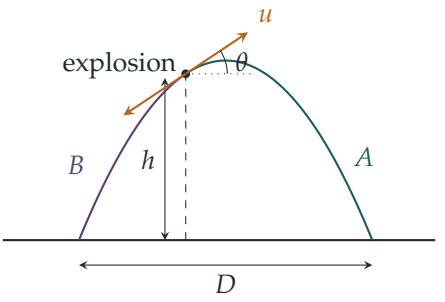
Solution

Setting up

We model the shell and the fragments as particles moving under gravity alone. The shell rises with initial speed v, so it comes to rest at height

$$h = \frac{v^2}{2g}, \qquad \text{that is,} \qquad 2gh = v^2.$$

Just after the explosion, call the fragments A and B. Their velocities are equal and opposite, so both lie in one vertical plane through the explosion point. Take x horizontal in that plane and y vertically upwards, with the explosion point at $(0, h)$. We can label the fragments so that A does not move downwards. Then A has velocity $(u\cos\theta,\ u\sin\theta)$ and B has velocity $(-u\cos\theta,\ -u\sin\theta)$, with $0 \le \theta \le \frac{\pi}{2}$. Write $s = \sin\theta$ and $c = \cos\theta$, so $c \ge 0$.



Landing times

Fragment A is at height $h + ust - \frac{1}{2}gt^2$ at time t after the explosion. It lands when

$$\tfrac{1}{2}gt^2 - us\,t - h = 0.$$

The product of the roots is $-2h/g < 0$, so there is exactly one positive root:

$$t_A = \frac{us + \sqrt{u^2s^2 + 2gh}}{g} = \frac{us + \sqrt{u^2s^2 + v^2}}{g}.$$

For B the vertical velocity is $-us$, so we replace s by $-s$:

$$t_B = \frac{-us + \sqrt{u^2s^2 + v^2}}{g}.$$

This is positive, since $\sqrt{u^2s^2 + v^2} > us$.

The distance between the landing points

Horizontally, A travels $uc\,t_A$ in the positive x-direction and B travels $uc\,t_B$ in the negative direction, so they land on opposite sides of the launch point. The distance between the landing points is therefore

$$D = uc\,(t_A + t_B) = \frac{2uc\sqrt{u^2s^2 + v^2}}{g}.$$

Since $D \geq 0$, maximising D is the same as maximising D^2. Put $x = s^2 \in [0,1]$, so that $c^2 = 1 - x$. Then

$$\frac{g^2D^2}{4u^2} = (1-x)(v^2 + u^2x) = f(x), \qquad \text{say,}$$

and

$$f(x) = v^2 + (u^2 - v^2)x - u^2x^2.$$

This is a quadratic in x with negative leading coefficient, and its vertex is at

$$x^* = \frac{u^2 - v^2}{2u^2}.$$

Ask yourself The vertex of f is a maximum, so why isn't the answer always $f(x^*)$?

Only $x \in [0,1]$ is allowed, because $x = \sin^2\theta$. Note $x^* < \frac{1}{2}$ always, so the vertex never lies beyond the top end. Whether the vertex is allowed depends only on its sign.

Case $u \le v$. Here $x^* \le 0$. On $[0,1]$ we are to the right of the vertex of a downward parabola, so f is decreasing there, and the maximum is at $x = 0$, where $f(0) = v^2$. This corresponds to $\theta = 0$, a horizontal explosion. Hence

$$D_{\max} = \frac{2u}{g}\sqrt{v^2} = \frac{2uv}{g}.$$

Case $v \le u$. Here $0 \le x^* < \frac{1}{2}$, so the vertex is allowed and gives the maximum:

$$\begin{aligned} f(x^*) &= v^2 + \frac{(u^2-v^2)^2}{2u^2} - \frac{(u^2-v^2)^2}{4u^2} = v^2 + \frac{(u^2-v^2)^2}{4u^2} \\ &= \frac{4u^2v^2 + u^4 - 2u^2v^2 + v^4}{4u^2} = \frac{(u^2+v^2)^2}{4u^2}. \end{aligned}$$

Therefore

$$D_{\max} = \frac{2u}{g}\cdot\frac{u^2+v^2}{2u} = \frac{u^2+v^2}{g}.$$

This happens when $\sin^2\theta = \dfrac{u^2-v^2}{2u^2}$, so the fragments fly off at an angle to the horizontal.

When $u = v$ both formulas give $2u^2/g$, so the two answers agree on the boundary between the cases, as they must. This completes the proof.

Checks

- Units: uv/g and $(u^2+v^2)/g$ are both (speed)^2/(acceleration), which is a length.
- Taking $g = 9.8$, $u = 30$ and $v = 10$, the best angle is $\theta \approx 41.8°$, and $D = 1000/9.8 \approx 102.04\,\text{m}$ from both the general formula for D and $(u^2+v^2)/g$. A horizontal explosion gives only $2\cdot 30\cdot 10/9.8 \approx 61.2\,\text{m}$.
- Stepping the two trajectories numerically and scanning θ over a fine grid reproduces both maxima for hundreds of random pairs (u,v).
- With $v = 0$ the shell explodes at ground level, and the answer becomes u^2/g. That is the usual maximum range of a projectile launched at speed u at 45°, as it should be: B goes into the ground at once, and $x^* = \frac{1}{2}$ gives $\theta = 45°$.

Common slips

- Assuming the best explosion is horizontal. That is only true when $u \le v$.

- Using the same time of flight for both fragments. The one thrown upwards stays in the air longer.
- Forgetting the constraint $0 \leq \sin^2\theta \leq 1$ when maximising, and so missing the case split.

Beyond STEP

The tidy answer has a neat explanation. Relative to the centre of mass, which simply falls straight down, the fragments move apart in a straight line at relative speed $2u$. But they hit the ground at different times, so the problem becomes one of optimising under a constraint, with a maximum that switches from the boundary to the interior as a parameter changes. This kind of switch reappears in the Lagrange multiplier and Karush–Kuhn–Tucker conditions of first-year optimisation. For conservation of momentum, centre-of-mass frames and projectile problems taken further, see Kleppner and Kolenkow, *An Introduction to Mechanics*.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.